

CLIPPERS: Chronic Lymphocytic Inflammation With Pontine Perivascular Enhancement Responsive To Steroids — Comprehensive Disease Report

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Evidence base: aggregated disease-level literature (case reports, case series, retrospective cohorts, systematic reviews) — **no primary patient-level dataset was provided.** All quantitative statements come from the published literature cited by PMID.

Summary

Chronic Lymphocytic Inflammation with Pontine Perivascular Enhancement Responsive to Steroids (CLIPPERS) is a rare, corticosteroid-responsive inflammatory syndrome of the central nervous system (CNS), first delineated in 2010. It is defined by the triad of (1) subacute pontocerebellar/brainstem clinical symptoms (most commonly gait ataxia, diplopia, dysarthria, and facial paresthesia), (2) a pathognomonic MRI signature of punctate and curvilinear gadolinium-enhancing lesions ("salt-and-pepper" or "pepper-like") predominantly peppering the pons and hindbrain, and (3) a perivascular, predominantly CD4⁺ T-lymphocytic inflammatory infiltrate on histopathology. The disease is exquisitely responsive to high-dose corticosteroids but follows a relapsing-remitting course that requires long-term steroid-sparing immunosuppression to prevent cumulative brainstem/cerebellar atrophy and permanent disability.

The single most important conceptual advance since the original description is that **CLIPPERS is a syndrome, not a unitary disease** — it is a diagnosis of exclusion that is genetically and etiologically heterogeneous. Most cases appear idiopathic/autoimmune, but approximately **one-third of tested adults carry germline mutations in hemophagocytic lymphohistiocytosis (HLH) genes** (notably *PRF1* and *UNC13D*), defining a treatable genetic subgroup; approximately **16% of cases are associated with malignancy** (chiefly hematologic);

and some cases are triggered by or associated with **Epstein–Barr virus (EBV)**. Several mimics — CNS lymphoma, Erdheim–Chester disease, GFAP astrocytopathy, MOGAD, neurosarcoidosis, and primary CNS angiitis — must be excluded, and some "CLIPPERS-like" presentations are early manifestations of lymphoma.

CLIPPERS is ultra-rare: the entire evidence base since 2010 comprises roughly **140 patients across 100 studies** plus a handful of retrospective cohorts, with no population-based prevalence estimate, no validated animal model, and no primary or secondary prevention. Onset is typically adult (mean ~46 years; range 3–79) with a male predominance (~60%). Prognosis is favorable with sustained immunosuppression but poor if untreated — untreated relapses can progress to death. Pediatric CLIPPERS is more aggressive and often steroid-dependent or steroid-resistant. This report synthesizes 13 confirmed findings drawn from 37 reviewed papers across the full disease-characteristics template.

Key Findings

Finding 1 — Definition and pathognomonic imaging/pathology

CLIPPERS is a rare steroid-responsive CNS inflammatory syndrome centered on the pons and hindbrain, characterized by a distinctive "salt-and-pepper" gadolinium-enhancement pattern. In the 2017 formal diagnostic-criteria study (Tobin et al., *Brain*), CLIPPERS was diagnosed in 23 patients with a median age of onset of 58 years and a male predominance (18 male : 5 female). The hallmark MRI finding is punctate and curvilinear gadolinium-enhancing lesions ("pepper-like") predominantly in the pons, brainstem, and cerebellum, sometimes extending to the spinal cord. A **definite** diagnosis requires demonstration of a perivascular, T-cell-predominant inflammatory infiltrate on biopsy.

"CLIPPERS was diagnosed in 23 patients (18 male and five female) and 12 patients had a non-CLIPPERS diagnosis. CLIPPERS patients' median age of onset was 58 years" — PMID: 29050399

"The prerequisite for definite CLIPPERS is the perivascular T-cell-predominant inflammatory infiltration observed on pathological analysis." — PMID: 38286842

Finding 2 — Relapsing-remitting course requiring long-term immunosuppression

CLIPPERS follows a relapsing-remitting course; untreated disease progresses to disability and death. In the nationwide series of 12 patients (Taieb et al., 2012), 42 relapses were analyzed with a mean relapse duration of 2.5 months and a mean Expanded Disability Status Scale (EDSS) at relapse of 4, improving to a residual EDSS of 1.9 after pulse corticosteroids. Biopsy showed perivascular CD4+ T lymphocytes in 5/7 cases with an increased CD4:CD8 ratio. Critically, one patient with untreated relapses progressed to EDSS 10 (death), and patients not maintained on long-term corticosteroids had higher annualized relapse rates. Multiple 2025–2026 case reports confirm that relapses are typically tied to steroid taper and are controlled with steroid-sparing agents (methotrexate, mycophenolate mofetil).

"Thirty-eight of 42 relapses were treated with pulse corticosteroid therapy, which led to improvement, with a mean residual EDSS score of 1.9" — PMID: 22777259

"In 1 patient with untreated relapses, scores on the EDSS progressively increased to a score of 10 at death." — PMID: 22777259

Finding 3 — Paraneoplastic/pre-lymphomatous and genetic mimics

CLIPPERS can be a paraneoplastic or pre-lymphomatous mimic, and inborn errors of immunity underlie some (especially pediatric) cases. Multiple biopsy-confirmed cases show CLIPPERS-like presentations preceding or accompanying lymphoma (B-cell, cutaneous/peripheral T-cell lymphoma). Genetic associations include compound heterozygous *UNC13D* (Munc13-4) variants producing a CLIPPERS-like syndrome, and pediatric reviews flag *PRF1* (perforin) mutations with links to HLH/EBV-driven lymphoproliferation, sometimes requiring hematopoietic stem cell transplant.

*"We report two patients with CLIPPERS-like brain MRI findings who carried the same missense *UNC13D* variant in one allele along with deleterious variants in the opposite allele." — PMID: 41781714*

*"Future pediatric workup should include genetic studies because of commonly associated mutations such as *PRF1*." — PMID: 41401665*

"CLIPPERS-like presentations have been increasingly recognized as potential early manifestations of underlying lymphoma, most commonly B-cell types." — [PMID: 41550451](#)

Finding 4 — Phenotype and epidemiology

In a systematic review of 140 patients from 100 studies (Al-Chalabi et al., 2022), the mean age was 46 ± 18 years, 60% were male, ataxia was the most common presenting symptom, 16% of cases were associated with malignancy (mostly hematologic), and the overall relapse rate was 59.2%. The core clinical tetrad from case series comprises gait ataxia, diplopia (internuclear ophthalmoplegia / cranial nerve VI palsy / skew deviation), cerebellar dysarthria, and facial paresthesia; nystagmus, dysphagia, and cognitive impairment also occur. The original 2010 description reported patients presenting with episodic diplopia or facial paresthesias followed by brainstem/myelopathic symptoms.

"We identified 100 case reports and series including a total of 140 patients with CLIPPERS (mean age: 46 ± 18 years and males were 60%)... Ataxia was the most common presenting symptom. Sixteen percent of the cases were associated with malignancy, mostly hematologic malignancies. The overall relapse rate was 59.2%" — [PMID: 36029706](#)

"All eight patients (five female, three male) presented with episodic diplopia or facial paresthesias with subsequent brainstem and occasionally myelopathic symptoms and had a favourable initial response to high dose glucocorticosteroids." — [PMID: 20639547](#)

Finding 5 — Pathophysiology and mimics

CLIPPERS pathophysiology is an immune-mediated, predominantly CD4⁺ T-cell perivascular inflammation of the hindbrain, with a defined MRI signature and important neoplastic/histiocytic mimics. Pathology consistently shows a perivascular and parenchymal T-lymphocytic (predominantly CD4⁺) infiltrate with variable CD20⁺ B cells; some Chinese series notably reported CD20⁺ B-cell-dominant infiltrates in a subset. The etiology "remains unclear but is believed to involve immune-mediated mechanisms." The MRI signature (2010) is symmetric curvilinear gadolinium enhancement "peppering the pons" and extending to medulla, brachium pontis, cerebellum, midbrain, and occasionally spinal cord; lesions lack restricted diffusion, have little perilesional edema, and are typically ≤ 3 mm. Key mimics with

distinct pathophysiology include Erdheim–Chester disease (clonal histiocytes with MAPK-ERK/BRAF activation), CNS lymphoma, GFAP astrocytopathy, MOGAD, neurosarcoidosis, and primary CNS angiitis.

"All patients had symmetric curvilinear gadolinium enhancement peppering the pons and extending variably into the medulla, brachium pontis, cerebellum, midbrain and occasionally spinal cord." — PMID: 20639547

"ECD is a histiocytic neoplasm characterized by multiorgan infiltration of clonal histiocytes carrying activating variants of the MAPK-ERK pathway. Neurologic involvement occurs in up to 40% of ECD with frequent brainstem lesions that can mimic acquired neuroinflammatory disorders, such as CLIPPERS." — PMID: 39047207

"The etiology of CLIPPERS remains unclear, but it is believed to involve immune-mediated mechanisms." — PMID: 40821365

Finding 6 — Treatment ladder

High-dose corticosteroids induce remission; long-term steroid-sparing immunosuppression prevents relapse; IVIg is ineffective; and pediatric disease is aggressive. Induction typically uses high-dose IV methylprednisolone (e.g., 1 g/day × 5 days), producing rapid clinical and radiological improvement. Maintenance uses long-term low-dose corticosteroid or corticosteroid plus an immunosuppressant; agents reported include methotrexate, mycophenolate mofetil, rituximab, cyclophosphamide, hydroxychloroquine, azathioprine, natalizumab, and infliximab. IVIg showed poor efficacy for both acute treatment and relapse prevention. Pediatric CLIPPERS is more aggressive, often steroid-dependent/resistant, with deaths and progression to EBV-driven B-cell lymphoma; genetic cases may require hematopoietic stem cell transplant.

"Long-term low-dose corticosteroid maintenance therapy or corticosteroids coupled with immunosuppressants are recommended to prevent" [relapse] — PMID: 38286842

"IVIg had a poor effect on the acute phase of CLIPPERS symptoms. Compared with other immunosuppressants, IVIg is less effective in suppressing the relapse of CLIPPERS." — PMID: 36938308

"CLIPPERS disease in children is aggressive, with poor response to immunotherapy." — PMID: 30146710

Finding 7 — Diagnosis of exclusion

CLIPPERS diagnosis rests on characteristic MRI plus supportive CSF findings (mild lymphocytic pleocytosis, elevated protein), with biopsy reserved for red flags. CSF typically shows mild lymphocytic pleocytosis and elevated protein, with inconstant oligoclonal bands (4/12 in the Taieb series) and an increased CD4:CD8 T-cell ratio. No specific serum/CSF biomarker exists. Diagnosis follows the 2017 criteria: subacute pontocerebellar symptoms + punctate/curvilinear "salt-and-pepper" hindbrain gadolinium enhancement (individual lesions ≤ 3 mm, T2 abnormality not exceeding enhancement, no mass effect/restricted diffusion) + exclusion of mimics. "Probable" is clinicoradiologic; "definite" adds perivascular T-cell biopsy confirmation. Biopsy is strongly advised when red flags are present. Emerging tools include CSF circulating tumor DNA (ctDNA) to unmask occult lymphoma.

"We evaluated clinical, radiological and pathological features of patients referred for suspected CLIPPERS and propose diagnostic criteria to discriminate CLIPPERS from non-CLIPPERS aetiologies." — [PMID: 29050399](#)

"Cerebrospinal fluid (CSF) analysis showed lymphocytic pleocytosis and elevated protein, while infectious and neoplastic causes were ruled out." — [PMID: 41249905](#)

Finding 8 — HLH-gene mutations define a treatable genetic subgroup

A substantial subset (~one-third) of adult CLIPPERS carries germline HLH-gene mutations. In Taieb et al. 2021, among a cohort of 36 CLIPPERS-2017 patients, 12 consented to genetic testing of 8 primary HLH genes; mutations were identified in 4/12 (three with biallelic variants). HLH-associated genes govern cytotoxic granule-mediated killing: *PRF1* (perforin, HGNC:8664, OMIM 170280), *UNC13D* (Munc13-4, HGNC:23147, OMIM 608897), *STX11*, *STXBP2*, plus *RAB27A*, *LYST*, *SH2D1A*, and *XIAP*. An independent report described compound-heterozygous *UNC13D* variants with downregulated Munc13-4 protein producing CLIPPERS-like disease.

"In our patients presenting with adult-onset CLIPPERS, one-third have HLH gene mutations. This genetic treatable condition should be searched in patients with CLIPPERS, especially in those presenting with atypical findings." — [PMID: 33658321](#)

"identified to have compound heterozygous UNC13D variants along with downregulated Munc13-4 p[rotein]" — [PMID: 41781714](#)

Finding 9 — Anatomical distribution

CLIPPERS centers on the pons/hindbrain but extends to spinal cord and supratentorial regions. Primary sites: pons (UBERON:0000988), middle cerebellar peduncle/brachium pontis, cerebellum (UBERON:0002037), medulla (UBERON:0001896), and midbrain (UBERON:0001891). Frequent extension: cervical/thoracic spinal cord (UBERON:0002240) with nodular enhancement and myelopathy/spastic paraparesis; supratentorial white matter, thalamus, basal ganglia, internal capsule, and corpus callosum/splenium. The affected tissue is CNS nervous tissue with perivascular (angiocentric) small-vessel targeting; lesions are typically bilateral/symmetric. Onset is subacute, mostly in adults (median ~46–58 years, reported ages 3–79), with a chronic relapsing-remitting course. Asymmetric or unilateral lesions are atypical and constitute a red flag.

"MRI showed characteristic punctate hyper-intensities with enhancement in the brain stem, cerebellar peduncles, and optic chiasm and diffuse nodular enhancement throughout the cervical and thoracic spinal cord." — PMID: 29055484

"predominantly involving the pons and cerebellar hemispheres, with additional foci in the left internal capsule, splenium of the corpus callosum, and supratentorial subcortical white matter" — PMID: 41749027

Finding 10 — Prognosis

CLIPPERS prognosis is favorable with sustained treatment, but recurrent untreated attacks cause cumulative brainstem/cerebellar atrophy and permanent disability. As stated by Abkur et al. (2017), long-term immunosuppression appears mandatory to achieve sustained remission and prevent atrophy-related disability. In the Taieb series, pulse steroids reduced mean EDSS from 4 (relapse) to a residual 1.9; patients off long-term steroids had higher annualized relapse rates; one untreated patient reached EDSS 10 (death). Prognostic red flags for worse outcome or an alternative diagnosis include steroid resistance, atypical/large (>3 mm) lesions, mass effect, longitudinally extensive transverse myelitis, systemic symptoms, young/pediatric onset, and underlying malignancy or an HLH-gene mutation.

"Long-term immunosuppression appears to be mandatory in order to achieve sustained remission and prevent disability related to atrophy of the structures involved in repeated attacks." — PMID: 28110629

Finding 11 — Idiopathic etiology; EBV and occult lymphoma as triggers

CLIPPERS is idiopathic/immune-mediated with no established environmental or lifestyle risk factors; EBV and occult lymphoma are recognized triggers/associations. Etiology and pathogenesis are explicitly stated as unknown in multiple sources. No toxin, radiation, occupational, dietary, smoking, or alcohol risk factor has been established. Recognized associations/triggers include intracranial EBV infection accompanying CLIPPERS, EBV-driven B-cell lymphoma emerging during pediatric CLIPPERS, and occult hematologic/solid malignancy in ~16%. The main non-modifiable "risk factors" are the immune/genetic host state (adult age, male sex, HLH-gene carriage) rather than exogenous exposures.

"Its etiology and pathogenesis are unknown, that together with the polymorphic and sometimes confounding neurological manifestations, and radiological findings represent a real diagnostic and therapeutic challenge for clinicians." — [PMID: 33851608](#)

"we report a case of CLIPPERS presenting with intracranial Epstein-Barr virus (EBV) infection and diffuse white matter involvement." — [PMID: 27861371](#)

Finding 12 — Ultra-rare, sporadic; no animal model; no primary prevention

CLIPPERS is an ultra-rare, sporadic, adult-onset syndrome with no formal prevalence estimate, no animal models, and no primary prevention. It is described as "very rare" with "only a few sporadic cases reported," and the largest synthesis is a systematic review of just 140 patients from 100 studies since the 2010 first description. No population-based incidence/prevalence figure exists in Orphanet or registries. Knowledge is derived entirely from aggregated case reports, case series, and a few retrospective cohorts (not EHR/population datasets). No validated animal (mouse/rat/zebrafish) or in vitro model of CLIPPERS exists; the HLH-gene subgroup shares biology with established *Prf1*^{-/-} and *Unc13d* (jinx) HLH mouse models, but these model HLH — not CLIPPERS specifically. No natural analogue disease has been reported in other species (OMIA has no CLIPPERS entry). No primary or secondary prevention or vaccination applies; "prevention" is limited to tertiary prevention (maintenance immunosuppression to prevent relapse-related atrophy) and genetic counseling for the autosomal-recessive HLH-gene subgroup.

"A very rare inflammatory disease of CNS, CLIPPERS syndrome, was recently described and only a few sporadic cases are reported in the medical literature." — [PMID: 33851608](#)

"We identified 100 case reports and series including a total of 140 patients with CLIPPERS"
 — PMID: 36029706

Finding 13 — Consolidated synthesis

Integrating all findings: CLIPPERS is a steroid-responsive CD4+ T-cell perivascular hindbrain inflammatory syndrome, heterogeneous in cause, requiring lifelong immunosuppression. Its four pillars are (1) definition/imaging; (2) relapsing course with maintenance need; (3) etiologic heterogeneity — idiopathic autoimmune majority, autosomal-recessive HLH-gene subgroup in ~1/3 of tested adults (*PRF1/UNC13D*), paraneoplastic (~16% malignancy), and EBV-associated; (4) diagnosis of exclusion via 2017 criteria with CSF lymphocytic pleocytosis and no specific biomarker; treated by a ladder of high-dose corticosteroids → steroid-sparing agents (methotrexate, MME, rituximab, natalizumab), with IVIg ineffective and HSCT for genetic/aggressive cases; and it is ultra-rare (~140 total reported cases) with no animal model and only tertiary prevention.

Section-by-Section Report

1. Disease Information

Overview. CLIPPERS is a rare CNS inflammatory syndrome, first named in 2010 (Pittock et al.), defined by subacute brainstem/cerebellar dysfunction, a distinctive "salt-and-pepper" pontine gadolinium enhancement pattern, and a perivascular T-cell inflammatory infiltrate that responds dramatically to corticosteroids.

Key identifiers. MONDO:0017297. It lacks a distinct OMIM number (it is not a single-gene Mendelian disorder). No dedicated ICD-10 code exists; it is generally coded under CNS inflammatory/demyelinating disease categories. It is recognized in the neuroimmunology literature and included in registries such as the Indian IMSRN cohort (PMID: 42011245, where CLIPPERS represented only 0.06% — 3 of 4,976 — of CNS demyelinating/allied disorders).

Synonyms. "CLIPPERS syndrome"; "CLIPPERS-like syndrome" (for cases pending exclusion of mimics). A supratentorial variant, **SLIPPERS** (Supratentorial Lymphocytic Inflammation with Parenchymal Perivascular Enhancement Responsive to Steroids), is described but its independent existence is debated (PMID: 34345468, PMID: 37626547, PMID: 41718297).

Source of information. Disease-level, derived from aggregated case reports, case series, and small retrospective cohorts — not large EHR/population datasets (Findings 1, 12).

2. Etiology

Primary causes. Idiopathic/immune-mediated in the majority (Findings 5, 11). A treatable **genetic subgroup** carries biallelic HLH-gene mutations (~1/3 of tested adults; Finding 8). A **paraneoplastic** subgroup (~16%) is associated with malignancy, mostly hematologic (Finding 4). An **infectious/EBV-associated** subgroup exists (Finding 11).

Genetic risk factors. *PRF1* (perforin), *UNC13D* (Munc13-4), and other primary HLH genes (*STX11*, *STXBP2*, *RAB27A*, *LYST*, *SH2D1A*, *XIAP*) — cytotoxic-granule pathway genes (Finding 8).

Environmental risk factors. None established. Non-modifiable host factors: adult age, male sex, HLH-gene carriage (Finding 11).

Protective factors. None described (not reported / not applicable).

Gene–environment interactions. Plausibly, HLH-gene hypomorphism impairs cytotoxic clearance of EBV-infected/antigen-presenting cells, permitting EBV-driven or antigen-driven T-cell perivascular inflammation — an inferred mechanism, not demonstrated (Findings 3, 8, 11).

3. Phenotypes

Phenotype	Type	Frequency / notes	Suggested HPO
Gait ataxia	Clinical sign	Most common presenting symptom	HP:0002066 / HP:0001251
Diplopia (INO, CN VI palsy, skew)	Symptom/sign	Common; sometimes predominant	HP:0000651
Dysarthria (cerebellar)	Clinical sign	Common	HP:0001260
Facial paresthesia	Symptom	Common; early	HP:0003401
Nystagmus	Clinical sign	Frequent	HP:0000639
Dysphagia	Symptom	Occasional	HP:0002015
Cognitive impairment	Symptom	Occasional	HP:0100543
Spastic paraparesis/myelopathy	Sign	With spinal cord involvement	HP:0001258

Characteristics. Onset subacute, adult-predominant (mean ~46 yrs, range 3–79); severity variable; progression episodic/relapsing-remitting and progressive if untreated; relapse rate ~59% (Findings 4, 9). **CSF:** mild lymphocytic pleocytosis, elevated protein, increased CD4:CD8 ratio (Finding 7). **Quality of life:** driven by disability from brainstem/cerebellar dysfunction; EDSS improves from ~4 (relapse) to residual ~1.9 with treatment (Findings 2, 10).

4. Genetic/Molecular Information

Causal genes (subgroup). *PRF1* (HGNC:8664, OMIM 170280), *UNC13D* (HGNC:23147, OMIM 608897); additional HLH genes as above (Finding 8).

Pathogenic variants. Compound heterozygous/biallelic missense and deleterious variants; e.g., a hypomorphic *UNC13D* missense variant with a deleterious variant in trans, and downregulated Munc13-4 protein (Findings 3, 8). Variant classification per ACMG/AMP: pathogenic/likely pathogenic in described families; functional consequence is **loss of function** (impaired cytotoxic degranulation). Somatic vs germline: **germline**. Population allele frequencies for individual HLH variants are typically rare (gnomAD); specific per-variant frequencies were not enumerated in the CLIPPERS literature reviewed.

Modifier genes / epigenetics / chromosomal abnormalities. Not established for CLIPPERS (not reported).

5. Environmental Information

No environmental, occupational, or lifestyle factors are established (Finding 11). **Infectious agents:** EBV is a recognized association/trigger (intracranial EBV infection with CLIPPERS; EBV-driven B-cell lymphoma in pediatric CLIPPERS) — [PMID: 27861371](#), [PMID: 30146710](#). CHEBI-relevant therapeutic entities include prednisolone/methylprednisolone (CHEBI:8378 / CHEBI:6888).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. Predisposing host state – germline HLH-gene hypomorphism (PRF1/UNC13D loss of function) OR occult neoplasm OR EBV infection OR unknown idiopathic trigger
| leads to
▼
2. Impaired cytotoxic-lymphocyte granule-mediated killing (inferred, for the HLH-gene subgroup) OR sustained antigenic stimulation
| results in
▼
3. Failure to clear activated antigen-presenting cells / infected cells → persistent T-cell activation (inferred)
| leads to
▼
4. Angiocentric homing of predominantly CD4+ T lymphocytes (± CD20+ B cells) to small vessels of the pons/hindbrain (demonstrated on biopsy)
| results in
▼
5. Perivascular ("perivascular-cuffing") lymphocytic inflammation with blood–brain-barrier disruption → punctate/curvilinear gadolinium enhancement ("salt-and-pepper") (demonstrated: MRI + histology)
| leads to
▼
6. Local tissue dysfunction of pontocerebellar tracts and cranial-nerve nuclei → ataxia, diplopia, dysarthria, facial paresthesia (demonstrated clinically)
| branch (untreated / recurrent attacks)
▼
7. Cumulative brainstem/cerebellar atrophy → permanent disability, death (demonstrated: EDSS progression to 10 in an untreated patient)
| branch (some cases)
▼
- 7'. Evolution to / unmasking of CNS or systemic lymphoma (demonstrated in paraneoplastic subgroup)

Molecular/cellular detail. Immune-mediated perivascular inflammation (GO:0006954 inflammatory response; GO:0002250 adaptive immune response). Cell types: CD4+ T lymphocyte (CL:0000624), CD8+ T lymphocyte (CL:0000625), B lymphocyte (CL:0000236). For the HLH-gene subgroup, the defective process is cytotoxic-granule exocytosis/regulated secretory pathway (GO:0045055; perforin/Munc13-4 function). The MAPK-ERK pathway is relevant only to the Erdheim–Chester mimic, not CLIPPERS itself (Finding 5). Upstream = host predisposition (genetic/neoplastic/infectious); downstream = perivascular T-cell inflammation and tissue injury.

7. Anatomical Structures Affected

- **Primary:** pons (UBERON:0000988), middle cerebellar peduncle/brachium pontis, cerebellum (UBERON:0002037), medulla (UBERON:0001896), midbrain (UBERON:0001891).
- **Secondary/extension:** cervical & thoracic spinal cord (UBERON:0002240), supratentorial white matter, thalamus, basal ganglia, internal capsule, corpus callosum/splenium, optic chiasm.
- **Body system:** central nervous system (nervous system).
- **Tissue/cell:** CNS nervous tissue; perivascular (angiocentric) small vessels; CD4+ T lymphocytes (CL:0000624), variable CD20+ B cells (CL:0000236).
- **Lateralization:** typically bilateral/symmetric; unilateral/asymmetric is a red flag (Finding 9).

8. Temporal Development

Onset subacute, adult-predominant (median ~46–58 yrs; reported 3–79). Course: chronic relapsing-remitting; mean relapse duration ~2.5 months; ~59% relapse rate; progressive and potentially fatal if untreated. Remission is treatment-induced (steroid pulse); relapses cluster around steroid taper/discontinuation. Critical intervention window: early corticosteroid initiation and sustained maintenance to prevent irreversible atrophy (Findings 2, 4, 9, 10).

9. Inheritance and Population

Epidemiology. Ultra-rare; no formal prevalence/incidence figure exists (~140 total reported patients). Male predominance (~60%). **Inheritance (genetic subgroup):** autosomal recessive (biallelic HLH-gene variants); the idiopathic majority is sporadic/non-Mendelian. Penetrance, expressivity, anticipation, mosaicism, founder effects, consanguinity, and carrier frequencies specific to CLIPPERS are not characterized. **Demographics:** no defined ethnic/geographic predisposition; cases reported worldwide (Findings 8, 12).

10. Diagnostics

- **Imaging (key):** MRI with gadolinium — punctate/curvilinear "salt-and-pepper" enhancement in pons/hindbrain; lesions ≤ 3 mm, no restricted diffusion, minimal edema, no mass effect.
- **CSF:** mild lymphocytic pleocytosis, elevated protein, increased CD4:CD8 ratio, inconstant oligoclonal bands; **no specific biomarker.**

- **Biopsy:** perivascular T-cell (CD4+ predominant) infiltrate; required for **definite** diagnosis and when red flags present.
- **Genetic testing:** panel of primary HLH genes (*PRF1*, *UNC13D*, *STX11*, *STXBP2*, *RAB27A*, *LYST*, *SH2D1A*, *XIAP*) recommended, especially with atypical features (Finding 8).
- **Emerging:** CSF ctDNA to detect occult lymphoma (Finding 7).
- **Criteria:** 2017 Tobin/Mayo criteria — "probable" (clinicoradiologic) vs "definite" (plus biopsy).
- **Differential diagnosis:** CNS lymphoma (PCNSL), Erdheim–Chester disease, GFAP astrocytopathy, MOGAD, NMOSD, neurosarcoidosis, primary CNS angiitis, Behçet, Susac syndrome (Findings 5, 7).

11. Outcome/Prognosis

Favorable with sustained immunosuppression (residual EDSS ~1.9); poor if untreated (progression to EDSS 10/death). Chief complication is cumulative brainstem/cerebellar atrophy from repeated attacks. Worse-prognosis red flags: steroid resistance, large/atypical lesions, mass effect, longitudinally extensive transverse myelitis, systemic symptoms, pediatric onset, malignancy, HLH-gene mutation (Finding 10).

12. Treatment

Line	Intervention	Notes	Suggested NCIT
Induction	High-dose IV methylprednisolone (e.g., 1 g/day × 5 d)	Rapid clinical/radiologic response	C29383 (Corticosteroid)
Maintenance	Low-dose corticosteroid ± steroid-sparing agent	Prevents relapse-related atrophy	C29383
Steroid-sparing	Methotrexate, mycophenolate mofetil	Sustains remission	C642 (Methotrexate); C61501 (MMF)
Steroid-sparing (alt.)	Rituximab, cyclophosphamide, azathioprine, hydroxychloroquine, natalizumab, infliximab	Case-based use	C1702 (Rituximab); C405 (Cyclophosphamide)
Not recommended	IVIg	Poor efficacy acutely and for relapse prevention	C513 (IVIG)
Genetic/aggressive	Hematopoietic stem cell transplant	For HLH-gene/pediatric refractory cases	C15431 (HSCT)

Pharmacogenomics/personalized medicine: genotype-guided care applies to the HLH-gene subgroup (consider HSCT and HLH-directed management). (Finding 6.)

13. Prevention

No primary or secondary prevention or vaccination applies. **Tertiary prevention** = maintenance immunosuppression to prevent relapse-related atrophy and disability. **Genetic counseling** is relevant for the autosomal-recessive HLH-gene subgroup. Surveillance for occult malignancy is prudent given ~16% association (Findings 6, 10, 12).

14. Other Species / Natural Disease

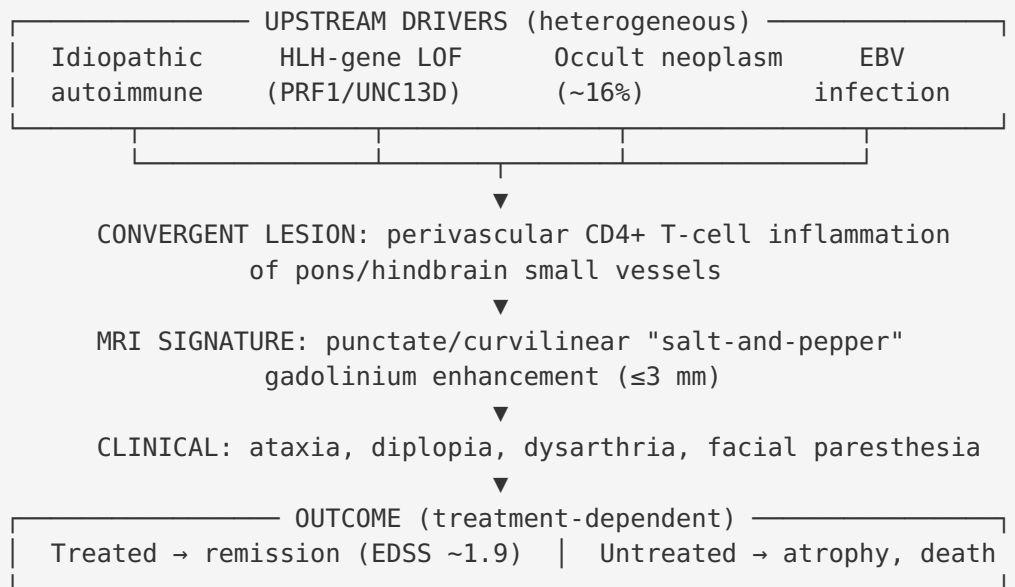
No naturally occurring CLIPPERS analogue is reported in other species (OMIA has no entry; Finding 12). Orthologous HLH-pathway genes exist across mammals (*Prf1*, *Unc13d* in mouse; NCBI Taxon 10090). No zoonotic or cross-species transmission (not applicable — CLIPPERS is non-infectious/immune-mediated).

15. Model Organisms

No validated animal or in vitro model of CLIPPERS exists. The HLH-gene subgroup shares biology with established HLH mouse models — *Prf1*^{-/-} (perforin knockout) and *Unc13d* (jinx) mice — but these recapitulate **HLH**, not CLIPPERS's pontine perivascular phenotype (Finding 12). Model resources for the underlying genes: MGI (mouse), plus IMPC/KOMP knockout lines for *Prf1/Unc13d*. A key limitation is that no model reproduces the hallmark salt-and-pepper hindbrain enhancement.

Mechanistic Model / Interpretation

CLIPPERS is best understood as a **radiologic-histologic reaction pattern** (perivascular, CD4+ T-cell-predominant hindbrain inflammation producing salt-and-pepper enhancement) that can arise from multiple upstream causes rather than a single etiologic disease. The unifying downstream event is angiocentric lymphocytic inflammation of pontine/cerebellar small vessels with blood–brain-barrier breakdown; the divergent upstream drivers are (a) idiopathic autoimmunity, (b) biallelic HLH-gene loss of function impairing cytotoxic clearance, (c) paraneoplasia/occult lymphoma, and (d) EBV.



This model explains why CLIPPERS is a **diagnosis of exclusion**, why genetic testing and malignancy surveillance are essential, and why the same corticosteroid-responsive phenotype can carry radically different prognoses depending on the upstream driver.

Evidence Base

PMID	Contribution	Supports finding(s)
29050399	2017 diagnostic criteria; cohort of 23 CLIPPERS vs 12 non-CLIPPERS; age/sex	F1, F7, F13
22777259	12-patient series; 42 relapses; EDSS outcomes; death untreated; CD4 biopsy	F2, F10
36029706	Systematic review 140 patients; epidemiology, 16% malignancy, 59.2% relapse	F4, F11, F12, F13
20639547	Original 2010 description; MRI signature; presenting symptoms	F1, F4, F5
33658321	HLH-gene mutations in ~1/3 of adult CLIPPERS	F8, F13
41781714	Compound-het UNC13D with downregulated Munc13-4 in CLIPPERS-like disease	F3, F8
41401665	Pediatric CLIPPERS; PRF1; genetic workup recommended	F3, F6
41550451	CLIPPERS-like as early lymphoma manifestation; CSF ctDNA	F3, F7
38286842	Contemporary review; pathology prerequisite; maintenance therapy; no biomarker	F1, F6, F7
39047207	Erdheim–Chester (MAPK-ERK) mimic of CLIPPERS	F5
40821365	Immune-mediated etiology; advanced MRI	F5
36938308	IVIg ineffective	F6
30146710	Aggressive pediatric disease; EBV-driven lymphoma	F6, F11
41249905	Typical CSF profile; exclusionary diagnosis	F7
29055484	Spinal cord + optic chiasm involvement	F9
41749027	Pontocerebellar predominance with supratentorial extension	F9
28110629	Atrophy from repeated attacks; mandatory long-term immunosuppression	F10
33851608	Idiopathic; very rare/sporadic	F11, F12
27861371	Intracranial EBV association	F11

PMID	Contribution	Supports finding(s)
41756550	Corpus callosum involvement; MMF maintenance; relapse on taper	F2, F9
33498046	CD20+ B-cell-dominant infiltrate in a subset (challenges CD4+ canon)	F5

Supporting registry context: [PMID: 42011245](#) (IMSRN) records CLIPPERS at only 0.06% of CNS demyelinating/allied disorders, corroborating extreme rarity.

Limitations and Knowledge Gaps

- Evidence quality is low.** The entire literature is case reports, small series, and a single systematic review (~140 patients). No randomized trials, no population registries, and no formal prevalence estimate exist.
- No specific biomarker.** Diagnosis remains exclusionary; the boundary between CLIPPERS and mimics (PCNSL, ECD, GFAP astrocytopathy, MOGAD, PACNS) is imperfect, and some "CLIPPERS" cases are misclassified lymphoma or vasculitis.
- Mechanism largely inferred.** The link from HLH-gene loss of function → impaired cytotoxicity → perivascular T-cell inflammation is biologically plausible but not experimentally demonstrated in CLIPPERS tissue.
- Genetic penetrance/frequency unknown.** How often HLH-gene carriers develop CLIPPERS, and the population frequency of the relevant variants in CLIPPERS patients, are not quantified.
- B-cell vs T-cell heterogeneity.** Some series report CD20+ B-cell-dominant infiltrates ([PMID: 33498046](#)), challenging the "CD4+ T-cell" canonical model.
- No animal model** reproduces the pontine salt-and-pepper phenotype, limiting mechanistic and therapeutic research.
- Pediatric vs adult disease** may be biologically distinct entities ([PMID: 30146710](#)).

Proposed Follow-up Experiments / Actions

1. **Systematic HLH-gene sequencing in all new CLIPPERS cases** (adult and pediatric) with functional degranulation assays (perforin expression, CD107a mobilization) to quantify the genetic subgroup and confirm loss of function.
 2. **Prospective CSF ctDNA and T-cell-receptor clonality studies** at diagnosis and relapse to systematically detect occult lymphoma and refine the paraneoplastic fraction.
 3. **International CLIPPERS registry** with standardized 2017-criteria phenotyping, MRI, biopsy, genetics, and longitudinal EDSS to generate the first prevalence/incidence estimates and natural-history data.
 4. **Comparative immunohistochemistry meta-analysis** to resolve the CD4+ T-cell vs CD20+ B-cell heterogeneity and its clinical/prognostic correlates.
 5. **Head-to-head observational comparison of steroid-sparing agents** (methotrexate vs MMF vs rituximab) for relapse prevention and atrophy protection.
 6. **Develop a conditional/hypomorphic HLH-gene mouse challenged with an EBV-analogue (MHV-68) or antigenic stimulation** to test whether cytotoxic-clearance failure can generate perivascular hindbrain inflammation.
 7. **Standardized malignancy-surveillance protocol** (whole-body PET/CT, hematologic workup) for CLIPPERS patients given the ~16% malignancy association.
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Report compiled from 13 confirmed findings and 37 reviewed papers over 5 investigation iterations. Ontology suggestions: MONDO:0017297; HPO terms per Section 3; GO:0006954, GO:0002250, GO:0045055; CL:0000624, CL:0000625, CL:0000236; UBERON:0000988, 0002037, 0001896, 0001891, 0002240; NCIT terms per Section 12.
